🇮🇳 BharatSetu — Bridging India's Governance Gap with Multi-Agent AI

Track: AI / Machine Learning — *Multi-agent AI system that collaborates or debates to reach decisions*

Team: BharatSetu | **GitHub:** github.com/Omkarpmath/BharatSetu-FINAL

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1. Problem Statement

India has **1.4 billion citizens** interacting with **4,000+ government schemes** across **39 ministries**, fragmented emergency services, and a legal system most cannot navigate. The fundamental problem is threefold:

1. **Language Barrier:** 65% of rural Indians cannot access English-only digital portals. India's 22 official languages create a massive digital divide where citizens cannot communicate with government services in their mother tongue.
2. **Fragmented Services:** A single citizen's problem (e.g., a farmer needing crop insurance after a flood) spans multiple departments — agriculture, finance, disaster management, health — with no unified interface. Citizens are forced to navigate 20+ different portals with different logins, processes, and languages.
3. **Emergency Response Gap:** In life-threatening situations, citizens must manually call multiple helpline numbers (Police: 100, Ambulance: 108, Fire: 101, Women Helpline: 181, Disaster: 1078). There is no single-touch multi-agency emergency dispatch system accessible in local languages.

BharatSetu solves this by deploying a **multi-agent AI system** where five specialized domain agents collaborate in real-time across all 22 Indian languages to deliver unified citizen services, emergency response, and scheme discovery — all through a single voice-first mobile interface.

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2. Solution Architecture

2.1 Council of Five — Multi-Agent AI System

BharatSetu deploys **five specialized AI agents**, each an expert in one governance domain. They independently reason, share context, and collaborate to serve the citizen:

Agent	Domain	Specialization
🏛️ **Nagarik Mitra**	Civic & Municipal	Pothole complaints, water supply, streetlights, garbage, birth/death certificates
🏥 **Swasthya Sahayak**	Health & Wellness	Symptoms, vaccinations (U-WIN), ABHA health ID, nearest hospital, Ayushman Bharat
📋 **Yojana Saathi**	Government Schemes	PM-KISAN, MGNREGA, ration cards, pensions, 800+ schemes with eligibility matching
💰 **Arthik Salahkar**	Finance & Banking	UPI fraud detection, Jan Dhan accounts, MUDRA loans, cyber scam reporting

⚖️ **Vidhi Sahayak**	Legal Aid & Rights	Zero FIR filing, NALSA free legal aid, RTI, consumer rights, domestic violence support

2.2 How Agents Collaborate (Not Just Route)

Unlike simple chatbot routing, our agents implement a **debate-and-converge** pattern:

```
Citizen Query: "My husband beats me and has taken all my money"
                                  │
                    ┌─────────────┼─────────────┐
                    ▼             ▼             ▼
             Vidhi Sahayak   Arthik Salahkar   Nagarik Mitra
             (Legal Aid)     (Financial)       (Civic)
                    │             │             │
                    ▼             ▼             ▼
             "File Zero FIR   "Freeze joint    "Contact local
              under DV Act,   accounts, block   women's cell
              call 181 Women  unauthorized      at nearest
              Helpline"       transactions"     police station"
                    │             │             │
                    └─────────────┼─────────────┘
                                  ▼
                        Multi-Agent Collaboration Card
                        (shows all consulted agents)
                                  │
                                  ▼
                        Vidhi Sahayak responds
                        (primary), with cross-agent
                        context from Finance + Civic
```

Technical Implementation:

- **Shared MCP Context:** Last 8 messages from ALL agents are passed as `sharedContext` to the active agent's LLM prompt, enabling seamless cross-domain awareness.
- **Phi-4 Multi-Agent Detection** (`/api/intelligence/multi-agent`): A separate Phi-4 model call detects when a query spans multiple domains and returns a `consulted` array with confidence scores for each relevant agent.
- **Automatic Handoff with Context Preservation:** When agent routing detects a better-fit agent, the citizen's message is replayed in the new agent's context with full conversation history, so the citizen never repeats themselves.

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3. AI/ML Technical Architecture

3.1 Multi-Model AI Pipeline

BharatSetu uses a **cascaded multi-model architecture** with intelligent fallbacks:

```
┌──────────────────────────────────────────────────────────────┐
│                    CITIZEN INPUT (22 Languages)              │
│         "मेरे गाँव में सड़क टूटी हुई है"                      │
└──────────────────┬──────────────────────────────────────────┘
```

```
                  │
                  ▼
┌──────────────────────────────────────────────────────────┐
│  LAYER 1: Azure Content Safety                           │
│  • Text moderation (violence, hate, self-harm, sexual)   │
│  • Blocks high-severity content before reaching any model │
│  • Non-blocking: timeout → pass-through (don't block user) │
└──────────────────────────────────────────────────────────┘
                  │
                  │
                  ▼
┌──────────────────────────────────────────────────────────┐
│  LAYER 2: Translation Pipeline                           │
│  Primary: Azure AI Translator (22 languages → English)   │
│  Fallback: MyMemory API (free, no signup)                │
│  Cache: In-memory translator cache (avoid duplicate calls) │
│  Output: English text for model classification           │
└──────────────────────────────────────────────────────────┘
                  │
                  │
                  ▼
┌──────────────────────────────────────────────────────────┐
│  LAYER 3: Agent Classification (Parallel)                │
│                                                          │
│  ┌──────────────────┐  ┌──────────────────┐             │
│  │  PASS 1: Phi-4   │  │  PASS 2: Local   │             │
│  │  (GitHub Models) │  │  TF-IDF + SVC    │             │
│  │                  │  │  (fallback)      │             │
│  │  • One-shot      │  │                  │             │
│  │    classification │  │  • Python        │             │
│  │  • 5s timeout    │  │    sklearn       │             │
│  │  • Routing cache │  │  • 3s timeout    │             │
│  └──────────┬───────┘  └──────────┬───────┘             │
│           │ winner              │ fallback              │
│           └──────────┬──────────┘                       │
│                      ▼                                   │
│  ┌──────────────────────────────┐                       │
│  │  PASS 3: Client-side Keywords │                      │
│  │  (last resort, regex-based)   │                      │
│  └──────────────────────────────┘                       │
└──────────────────────────────────────────────────────────┘
                  │
                  ▼
┌──────────────────────────────────────────────────────────┐
│  LAYER 4: Response Generation                            │
│                                                          │
│  Primary: Azure OpenAI (GPT-4o-mini)                     │
│    • Dual deployment round-robin (A/B) for 2x TPM        │
│    • Personalized system prompt per agent                │
│    • Citizen profile injection (Aadhaar-verified data)   │
│    • DIGIPIN location-aware responses                    │
│                                                          │
│  Fallback Chain:                                         │
│    Azure A (429) → Azure B → GitHub Models GPT-4.1-mini  │
```

```
│    → GPT-4o → Rich demo response (never fails)              │
│                                                              │
│  Post-GPT: Phi-4 late result harvesting                      │
│    (if Phi resolved during GPT execution, use for handoff)  │
└──────────────────────────────────────────────────────────────┘
```

3.2 Key AI/ML Innovations

A. Parallel Dual-Model Routing (Phi-4 + GPT)

- **Problem:** Single-model routing is slow (3-12s) and unreliable for Indian languages.
- **Solution:** Fire Phi-4 classification in parallel with GPT response generation. Phi has a 5-second early timeout — if it resolves first, its routing is applied immediately. If Phi is slow (cold start), GPT starts without waiting. After GPT completes, a 100ms "late harvest" checks if Phi resolved during GPT execution.
- **Result:** Routing is correct 95%+ of the time with zero added latency.

B. Script-Agnostic Keyword Layer

- **Problem:** Azure Translator may be unavailable, and Phi-4 works best on English.
- **Solution:** A pre-translation regex layer with 15+ rules catches high-confidence keywords in Devanagari, Bengali, Telugu, Tamil, Kannada, Malayalam, Gujarati, Punjabi, and Urdu scripts — before any API call is made.
- **Example:** `घरेलू हिंसा` (domestic violence) → `vidhi_sahayak` immediately, without translation.

C. Few-Shot Protection Layer

- **Problem:** Small models sometimes misroute (e.g., "ration card" to health instead of welfare schemes).
- **Solution:** A background GPT-4o verification call runs in parallel with the main response. It uses 12 carefully crafted few-shot examples covering Indian government edge cases. If it disagrees with Phi-4's routing, it wins.

D. Translation-Aware Routing Cache

- **Problem:** Repeated queries waste API calls and add latency.
- **Solution:** In-memory caches for both translation (Azure Translator) and routing (Phi-4) results. Identical queries hit cache with 0ms latency.

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4. AI-Powered Features

4.1 Dynamic Form Generation with LLM

Endpoint: `/api/generate-form`

When a citizen says "I want to apply for a loan" or "file a complaint," the AI:

1. Analyzes the user's intent and the active agent's domain
2. Generates a **context-specific form schema** with field names, labels, validation rules, required documents, and auto-fill mappings
3. Auto-fills known fields from the citizen's Aadhaar-verified profile
4. Returns a structured JSON form rendered instantly in the UI

Example: Saying "apply for KCC" in Yojana Saathi generates a Kisan Credit Card application form with fields for land area, crop type, bank details — pre-filled with the citizen's name, location, and income from their profile.

4.2 Vision-AI Document Analysis

Endpoint: `/api/vision-chat`

Citizens can photograph documents (ration cards, Aadhaar, land records) and the AI:

1. Analyzes the image using Azure Computer Vision / GPT-4o vision
2. Extracts text, document type, and relevant data
3. Routes the analysis to the appropriate agent with context

4.3 Voice-First Interface (22 Languages)

Endpoints: `/api/stt` (Speech-to-Text), `/api/voice` (Text-to-Speech)

- **STT:** Azure Speech Services with real-time streaming, supports all 22 scheduled languages
- **TTS:** Azure Neural TTS reads agent responses aloud in the citizen's native language
- **Auto-routing:** Voice input is classified by Phi-4 and routed to the correct agent before the response is even generated

4.4 AI Content Safety

Endpoint: `/api/content-safety`

Every citizen message passes through Azure Content Safety before reaching any LLM:

- Checks for violence, hate speech, self-harm, and sexual content
- High-severity content is blocked with a culturally sensitive response
- Non-blocking: timeout or service failure → pass-through (never blocks the user)

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5. SOS Emergency System

5.1 Architecture

```
┌──────────────────────────────────────────────────┐
│  CITIZEN: Long-press SOS (3 seconds)              │
│  • GPS coordinates captured                       │
│  • ISRO DIGIPIN grid code generated (4×4 algorithm)│
│  • Audio recording starts automatically           │
└──────────────────────────────────────────────────┘
                   │
                   │ POST /api/sos
                   ▼
┌──────────────────────────────────────────────────┐
│  SOS ENGINE (sos-engine.ts)                       │
│  • Creates event with unique SOS ID               │
│  • Returns eventId immediately (async dispatch)   │
│  • Launches parallel fan-out to 7+ responders     │
└──────────────────────────────────────────────────┘
                   │
                   │ Async Fan-out
        ┌──────────┼──────────┬──────────┐
        ▼          ▼          ▼          ▼
    Police 100  Ambulance 108  Fire 101  Women 181
        │          │          │          │
        ▼          ▼          ▼          ▼
   ┌──────────────────────────────────────────┐
```

```
│  /api/sos/dispatch (per responder)          │
│  Priority: Real API → Webhook → Simulated   │
│  Webhook payload includes:                  │
│    eventId, GPS, DIGIPIN, responderType,    │
│    citizen message, timestamp               │
└─────────────────────────────────────────────┘
     │
     ▼
Client polls /api/sos/status for real-time
responder status updates (NOTIFIED / FAILED)
```

5.2 ISRO DIGIPIN Integration

- Uses the **official ISRO/India Post 4×4 grid subdivision algorithm** for location encoding
- Generates alphanumeric codes like `3PJ6-Q8P-84M` from GPS coordinates
- Enables emergency responders to locate the citizen even without precise addresses
- Look-up feature: Enter a DIGIPIN → get exact GPS coordinates and map location

5.3 WhatsApp SOS Alert

- Emergency alerts sent via WhatsApp to the configured emergency contact
- Includes GPS coordinates, DIGIPIN, and SOS event ID
- Works even with poor network (queued and sent when connectivity resumes)

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6. Tech Stack

Layer	Technology	Purpose
Frontend	Next.js 14 (React 18) + TypeScript	SSR, API routes, responsive mobile-first UI
State	Zustand	Client-side store for chat history, forms, SOS
Styling	Tailwind CSS	Glassmorphism, dark mode, micro-animations
AI Models	Azure OpenAI (GPT-4o-mini), GitHub Models (GPT-4.1-mini, Phi-4)	Response generation + routing
Translation	Azure AI Translator + MyMemory API	22 Indian languages ↔ English
Speech	Azure Speech Services	STT (22 languages) + Neural TTS
Safety	Azure Content Safety	Text moderation before LLM
Vision	Azure Computer Vision / GPT-4o Vision	Document and image analysis
Location	ISRO DIGIPIN Algorithm + Browser Geolocation	4×4 grid encoding for emergency dispatch
SOS	Custom async dispatch broker	Fan-out to Police, Ambulance, Fire, NDRF, Women Helpline, Cyber Crime, Legal Aid
i18n	Custom translation system	22 official languages with data-i18n attribute injection into iframes

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7. API Endpoints (17 Routes)

Route	Method	Description
`/api/agent`	POST	Multi-agent orchestration — routing + LLM response
`/api/intelligence/multi-agent`	POST	Phi-4 multi-domain collaboration detection
`/api/translate`	POST	Azure Translator with MyMemory fallback
`/api/generate-form`	POST	LLM-powered dynamic form schema generation
`/api/content-safety`	POST	Azure Content Safety moderation
`/api/vision-chat`	POST	Image/document analysis via AI vision
`/api/grievance`	POST	Civic complaint filing with Phi-4 ticket generation
`/api/schemes`	GET	Government scheme database search
`/api/health`	GET	Health facilities and info
`/api/stt`	POST	Speech-to-text (Azure Speech)
`/api/voice`	POST	Text-to-speech (Azure Neural TTS)
`/api/sos`	POST	Emergency SOS trigger — creates event + async dispatch
`/api/sos/dispatch`	POST	Single responder dispatch (webhook/API/simulated)
`/api/sos/status`	GET	Poll SOS event status and responder updates
`/api/sos/end`	POST	End active SOS session
`/api/sos/sms`	POST	SMS alerts via Fast2SMS / MSG91
`/api/sos/update-location`	POST	Real-time GPS location updates during SOS

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8. Multi-Agent Collaboration — Detailed Flow

8.1 Message Lifecycle

```
1. Citizen types/speaks: "मेरी पत्नी को आयुष्मान कार्ड चाहिए"
   (My wife needs an Ayushman card)

2. Content Safety Check → PASS

3. Translation: "My wife needs an Ayushman card"

4. Routing Decision:
   ├─ Phi-4 classifies → "yojana_saathi" (scheme, not health)
   ├─ Raw Override → No match
   ├─ Keyword Override → "ayushman card" → yojana_saathi ✓
   └─ Client Keywords → "ayushman" → swasthya_sahayak (WRONG)
```

```
   Winner: Phi-4 + Keyword agree → yojana_saathi

5. Context Assembly:
   ├─ System Prompt: Yojana Saathi role + citizen profile
   ├─ Shared Context: Last 8 msgs from Nagarik Mitra, Swasthya Sahayak
   ├─ Conversation History: Last 6 messages in current agent
   └─ DIGIPIN: Location-aware scheme suggestions

6. LLM Call: Azure OpenAI GPT-4o-mini
   ├─ Deploy A (round-robin) → 429? → Deploy B → 429? → GitHub GPT-4.1-mini
   └─ Response in Hindi (MANDATORY — citizen's chosen language)

7. Post-Response:
   ├─ Multi-Agent Detection: Phi-4 checks if Health should be consulted
   ├─ Auto-Track: Creates tracked item "Ayushman Card Application"
   └─ TTS: Speaks response in Hindi if enabled
```

8.2 Cross-Agent Context Sharing (MCP Pattern)

```
sharedContext = [
  "[Swasthya Sahayak] Citizen: मुझे बुखार है",
  "[Swasthya Sahayak] Agent: आपको नजदीकी PHC जाना चाहिए...",
  "[Arthik Salahkar] Citizen: मेरा UPI काम नहीं कर रहा",
  "[Arthik Salahkar] Agent: UPI server down है, 2 घंटे बाद try करें..."
]
```

This shared context is injected into every agent's system prompt, creating a **Model Context Protocol (MCP)** effect where agents are aware of each other's conversations without direct communication.

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9. Resilience & Failover Architecture

BharatSetu is designed to **never fail** — critical for a governance platform serving emergencies:

Failure Point	Fallback Chain
Azure OpenAI quota exhausted	→ Deploy B → GitHub GPT-4.1-mini → GPT-4o → Rich demo response
Phi-4 cold start (>5s)	→ Local TF-IDF classifier → Client-side keyword regex
Azure Translator unavailable	→ MyMemory free API → Pass-through (model handles raw script)
Content Safety timeout	→ Pass-through (don't block the user)
SOS webhook unreachable	→ Simulated dispatch with status tracking
SMS API not configured	→ Console log + graceful degradation
Network offline	→ Zustand client-side state preserves all data

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10. Alignment with Track 1: AI/ML

Track Bullet Point	BharatSetu Implementation
Multi-agent AI that collaborates or debates to reach decisions	✅ Council of Five agents with shared MCP context, Phi-4 multi-domain detection, and automatic cross-agent handoff
AI that critiques and improves its own reasoning loop	✅ Background GPT-4o route verification runs in parallel and overrides Phi-4 if it disagrees (self-correction)
Build a model that adapts in real-time based on user corrections	✅ Profile-based personalization — citizen's Aadhaar data, DIGIPIN, occupation, income calibrate every response
AI system that explains uncertainty in predictions	✅ Multi-Agent Collaboration Card shows confidence scores for each consulted agent
Detect hallucinations in AI outputs without ground truth	✅ Content Safety Layer + few-shot anchoring prevents false scheme information; keyword overrides ensure high-confidence routing

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11. Impact & Scalability

Current Capabilities

- **22 official Indian languages** — voice + text across all agents
- **7 emergency responder types** — parallel async dispatch
- **17 API routes** — comprehensive governance coverage
- **800+ government schemes** — AI-powered eligibility matching
- **3-tier AI fallback** — 99.9% uptime design

Future Roadmap

- Database integration (MongoDB/Supabase) for persistent citizen records
- Fine-tuning on actual government complaint data for improved routing
- Blockchain-based complaint tracking for transparency
- Real UMANG/DigiLocker API integration for document verification
- Offline-first PWA for rural areas with intermittent connectivity

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12. How to Run

```
# Clone the repository
git clone https://github.com/Omkarpmath/BharatSetu-FINAL.git
cd bharat_setu

# Install dependencies
npm install

# Configure environment variables
cp .env.example .env.local
# Add Azure OpenAI, GitHub Token, Azure Translator keys
```

```
# Start development server
npm run dev
# → http://localhost:3000
```

Required API Keys

Key	Service	Purpose
`GITHUB_TOKEN`	GitHub Models	GPT-4o / GPT-4.1-mini responses
`GITHUB_TOKEN_PHI`	GitHub Models	Phi-4 agent routing
`AZURE_OPENAI_API_KEY`	Azure OpenAI	Primary LLM (optional, GitHub is default)
`AZURE_TRANSLATOR_KEY`	Azure Translator	22-language translation
`AZURE_SPEECH_KEY`	Azure Speech	STT + TTS in Indian languages
`AZURE_CONTENT_SAFETY_KEY`	Azure Content Safety	Text moderation

> ***"BharatSetu is not just an app — it's a bridge between 1.4 billion citizens and their government, speaking every language India speaks."***